

Solution Requirements

Date	10/07/2026
Name	Jaswanth Madapurm
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements that define what the Credit Card Approval Prediction **System** should do and how effectively it should perform.

- **Functional Requirements (FR):** Define the specific features and operations of the Credit Card Approval Prediction System, such as user authentication, applicant data collection, data validation, machine learning prediction, result display, report generation, and secure data storage. These describe what the system should do.
- **Non-Functional Requirements (NFR):** Define the quality attributes of the Credit Card Approval Prediction System, including performance, accuracy, security, reliability, scalability, usability, and maintainability. These describe how well the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in to the application using valid credentials before accessing the prediction system.	High
2	Authorization Levels	Only authorized users (Admin/Bank Staff) can access applicant details, prediction results, and system reports.	High
3	External Interfaces	The system accepts applicant data through forms or CSV files and uses trained Machine Learning models for prediction.	High
4	Transaction Processing	The system validates applicant details, preprocesses data,	High

		predicts approval status, and stores the prediction result.	
5	Reporting	The system displays approval/rejection results and generates prediction summaries and reports.	Medium
6	Business Rules	Credit approval is based on factors such as income, employment status, credit history, loan amount, and debt ratio using the ML model.	High
7	Compliance to Laws or Regulations	Applicant data is securely stored and handled according to data privacy and banking guidelines.	High
8	Other	The system allows future retraining of the ML model using updated applicant datasets for improved prediction accuracy.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process applicant details and generate predictions quickly.	Prediction generated within 2 seconds
2	Scalability	The system should support increasing numbers of applicants without performance degradation.	Supports 5,000+ applications efficiently
3	Security & Data Privacy	Applicant information should be protected using secure authentication and encrypted storage.	Password encryption and secure data handling
4	Reliability & Availability	The prediction system should be available whenever users need it with minimal downtime.	99% system availability
5	Usability & Accessibility	The application should provide a simple, user-friendly interface that is easy to operate.	Easy navigation with clear input forms and prediction results
6	Other	The system should be maintainable and allow easy model updates with new datasets.	Supports model retraining and future enhancements

